

Homework 5

Due on April 7, 2025

Instructions:

- Install `pdflatex`, `R`, and `RStudio` on your computer.
- Please edit the `HW5_First_Last.Rnw` file in `Rstudio` and compile with `knitr` instead of `Sweave`. Go to the menu `RStudio|Preferences...|Sweave` choose the `knitr` option, i.e., `Weave Rnw files using knitr?` You may have to install `knitr` and other necessary packages.
- Replace "First" and "Last" in the file-name with your first and last names, respectively. Complete your assignment by modifying and/or adding necessary R-code in the text below.
- You should submit both a **PDF file** and a **ZIP file** containing the data and the `HW5_First_Last.Rnw` file required to produce the PDF. The file should be named "HW5_First_Last.zip" and it should contain a single folder named "First_Last" with all the necessary data files, the `HW5_First_Last.Rnw` and `HW5_First_Last.pdf` file, which was obtained from compiling `HW5_First_Last.Rnw` with `knitr` and `LATEX`.
- The GSI grader will annotate the PDF file you submit. However, they will also check if the code you provide in the ZIP file compiles correctly. If the file fails to compile due to errors other than missing packages, there will be an automatic 10% deduction to your score.

Problems:

1. Consider the Dow Jones data set provided in the file `DowJones30.csv` under the textbook data sets folder in Canvas. The data set is also available [here](#).
(a) Consider the vectors of daily log-returns of all 30 stocks

$$r_k, \quad k = 1, \dots, 2529$$

over the period of 10 years from 1990 through 2000. Model the joint distribution of the daily log-returns with a multivariate t -distribution with mean μ and covariance matrix Σ . More precisely,

$$r_k \stackrel{d}{=} \mu + T, \quad \text{with } T = \frac{X}{\sqrt{Y/\nu}},$$

where $X \sim \mathcal{N}(0, \Lambda = \Sigma \times (\nu - 2)/\nu)$ and $Y \sim \text{Gamma}(\nu/2, 1/2)$ are independent.

Fit this model to the data. Assuming that $\nu > 2$, estimate Λ by using the sample covariance of the data and use the method of profile likelihood to estimate ν .

```
library("mnormt")
t.loglike = function(dat, nu.range){
  mu = apply(dat, 2, mean);
  Sig = cov(dat);
  t.loglike = matrix(0, nrow=1, ncol=length(nu.range));
```

```

for (i in 1:length(nu.range)){
  nu = nu.range[i];
  t.loglike[1,i] = sum(log(dmt(dat,mean=mu, S = Sig/(nu/(nu-2)), df=nu)));
}
return(t.loglike);
}
dat = read.csv("./DowJones30.csv",head=T);
n = length(dat$AA);
x = unlist(dat);
x = matrix(as.numeric(x[-(1:n)]),nrow=n,ncol=30);
ret = log(x[-1,]/x[-n,]);
nu.range = seq(from=2.01,to=10,by=0.01);

loglike = t.loglike(ret, nu.range)
max_idx = which.max(loglike)
nu_MLE = nu.range[max_idx]
cat(nu_MLE)

## 7.25

```

Using the method above, we estimate that $\hat{\nu} = 7.25$.

(b) Using the method of profile likelihood, obtain an asymptotic 95%-confidence interval for the parameter ν .

```

alpha = 0.05
q.alpha = qchisq(1-alpha, 1)
z1 = (2*loglike > 2*max(loglike) - q.alpha)
ci_lb = min(nu.range[which(z1)])
ci_ub = max(nu.range[which(z1)])
cat(c(ci_lb, ci_ub))

## 6.92 7.6

```

The 95% confidence interval for ν is (6.92, 7.6).

(c) Compute the daily Value-at-Risk at level $\alpha = 0.05$ for the equal-weighted portfolio including all 30 stocks and using the model-fit in part (a), obtain $VaR_{0.95}$, so that

$$\mathbb{P}(r < -VaR_{0.95}) = 0.05,$$

where r denotes log-return on this portfolio.

```

mu = apply(ret,2,mean);
Lambda = cov(ret) / (nu_MLE/(nu_MLE-2))
d = 30;

```

```

w = rep(1/d, d);

VaR = function(VaR_level){
  VaR = -1 * sqrt(t(w) %*% (Lambda %*% w));
  VaR = VaR * qt(1-VaR_level, df = nu_MLE);
  VaR = VaR - w %*% mu;
  return(VaR)
}

cat(VaR(0.95))

## 0.01487999

```

Then, the Value-at-Risk at level $\alpha = 0.95$ is 0.01488 for the log-returns, which means that around 5% of the times, the portfolio has losses bigger than 1.5%.

2. Consider the data set from the previous problem. Assume that the vector of daily log-returns $r \stackrel{d}{=} \mu + T$, where $\mu = \bar{r}$ is the vector of sample means and T is distributed according to the t -distribution found in the previous problem.

(a) Given a target expected daily return level v , determine the portfolio weights w_i , $i = 1, \dots, 30$ such that $\sum_{i=1}^{30} w_i = 1$, which *minimize* $\text{VaR}_{0.95}$ of the portfolio return

$$r(w) := w_1 r_1 + \dots + w_{30} r_{30},$$

where r_i is the daily log-return of the i th stock in DJIA. Write an R -function that returns the optimal portfolio weights given a value for the target expected return.

How do the optimal portfolio weights for a given target expected return depend on the VaR level α ? Explain.

Consider $\sigma_w^2 := w^\top \Lambda w$. Then, by the hint, we know that:

$$\begin{aligned} r(w) &\stackrel{d}{=} w^\top \mu + (w^\top \Lambda w)^{1/2} T_\nu \\ &\stackrel{d}{=} w^\top \mu + \sigma_w T_\nu \end{aligned}$$

Thus, the Value-at-Risk must satisfy:

$$\begin{aligned} \mathbb{P}(r(w) < -\text{VaR}_\alpha) &= \mathbb{P}(w^\top \mu + \sigma_w T_\nu < -\text{VaR}_\alpha) \\ &= \mathbb{P}(T_\nu < \frac{-\text{VaR}_\alpha - w^\top \mu}{\sigma_w}) = 1 - \alpha \\ \Rightarrow \frac{-\text{VaR}_\alpha - w^\top \mu}{\sigma_w} &= q_{1-\alpha}(\nu) \\ \Rightarrow \text{VaR}_\alpha &= -w^\top \mu - (w^\top \Lambda w)^{1/2} q_{1-\alpha}(\nu) \end{aligned}$$

For a given target expected return, we have that $w^\top \mu = v$. Consequently, for a fixed VaR level, minimizing the Value-at-Risk is equivalent to minimizing the variance σ_w^2 . In conclusion, we have the minimization problem:

$$\begin{aligned} \min_w \quad & w^\top \Lambda w \\ \text{s.t.} \quad & w^\top \mu = v \\ & \sum_i w_i = 1 \end{aligned} \tag{1}$$

Additionally, since $\Lambda = \frac{\nu-2}{\nu} \Sigma$, we can change the objective function to $w^\top \Sigma w$. Then, the objective function becomes the portfolio variance.

```
library("quadprog");
mu = apply(ret,2,mean);
Sigma = cov(ret)
run_solve_QF <- function(Sigma, mu, v, short_sell = F) {
  d <- length(mu)
  Dmat <- 2 * Sigma # Required because solve.QP uses (1/2)x'Dx
  dvec <- rep(0, d)
  Amat <- cbind(mu, rep(1, d))
  bvec <- c(v, 1)
  if (short_sell){
    Amat = cbind(Amat, diag(d))
    bvec = c(bvec, rep(0, d))
  }
  res <- solve.QP(Dmat = Dmat,
                  dvec = dvec,
                  Amat = Amat,
                  bvec = bvec,
                  meq = 2)
  return(list(variance = res$value, solution = res$solution))
}
```

In conclusion, the minimization of Value-at-Risk depends on the minimization of a quadratic form, and the weights affect the VaR only through this quadratic form that is associated with the variance of the portfolio.

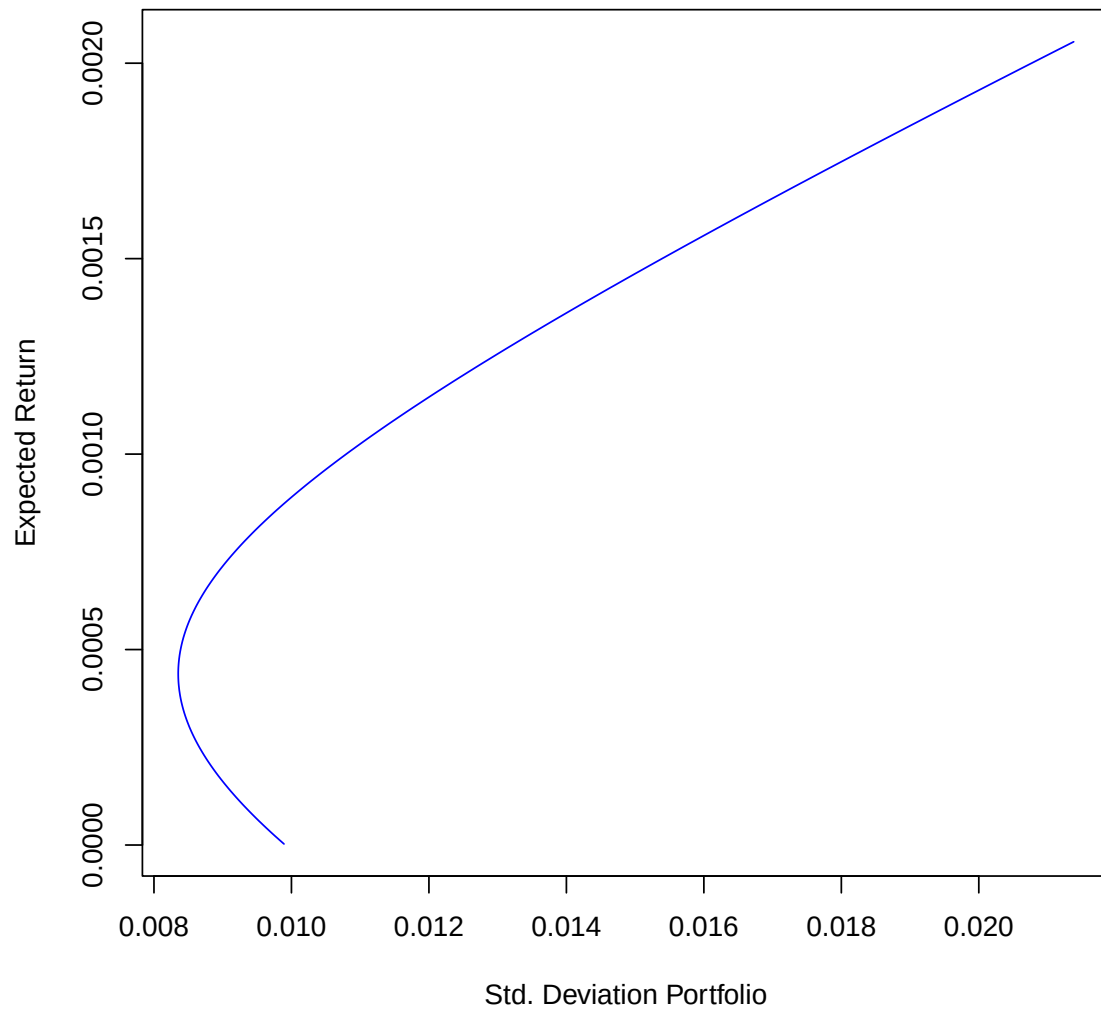
(b) Using the function in part (a) compute the efficient frontier of portfolios corresponding to several target daily log-return levels $v = 0.001/365$ to $v = 0.65/365$. Graph the points $(SD(r(w)), \mathbb{E}(r(w)))$, corresponding to these portfolios. That is, produce a risk-reward diagram for the portfolios, where $SD(r(w))$ and $\mathbb{E}(r(w))$ denote the standard deviation and expectation of the returns, respectively. Assuming that the **yearly** risk-free interest rate is $\mu_F = 0.03$, determine the “tangency” portfolio, i.e., the one maximizing the Sharpe ratio $(\mathbb{E}r(w) - \mu_F)/SD(r(w))$. (Note: remember to work in the correct units – i.e. determine the **daily** risk-free rate. You may also have to find an appropriate range of target expected returns in order to find the global maximum of the Sharpe ratio.)

```

v.range = seq(from=0.001,to=0.75,by=0.001)/365;
sig.pf = matrix(0,nrow=length(v.range),ncol=1);
weights = matrix(0,nrow=length(v.range),ncol=30);
for (i in (1:length(v.range))) {
  v = v.range[i];
  res = run_solve_QF(Sigma, mu, v, short_sell=F)
  sig.pf[i] = sqrt(res$variance);
  weights[i,] = res$solution;
}
mu_f = log(1.03)/365; # Risk-free rate
plot(sig.pf, v.range, type='l', col='blue',
      xlab="Std. Deviation Portfolio", ylab='Expected Return',
      main = "Risk-Reward Diagram")

```

Risk-Reward Diagram



```
sharp.ratio = (v.range - mu_f)/sig.pf
opt.sr.idx = which.max(sharp.ratio)
optimal.pf.short = c(v.range[opt.sr.idx], sig.pf[opt.sr.idx])
library(knitr)
kable(weights[opt.sr.idx, ])
```

x
0.1428250
-0.0019798
-0.3348414
-0.1563499
0.0297097
0.2052480
0.1046624
-0.0479539
-0.1594483
0.0936093
0.2489871
-0.1629834
0.0073034
0.1796263
0.1235208
0.1867875
-0.1405612
-0.2059089
0.0814058
0.1192858
0.0452893
0.1498127
0.1630413
0.1078004
0.0444390
-0.1218930
0.1637563
0.2163638
-0.0383044
-0.0432494

The column above represents the weights of the tangencial portfolio (i.e. maximizes Sharpe ratio). We can observe that some of the weights are negative. This indicates that short-selling is allowed.

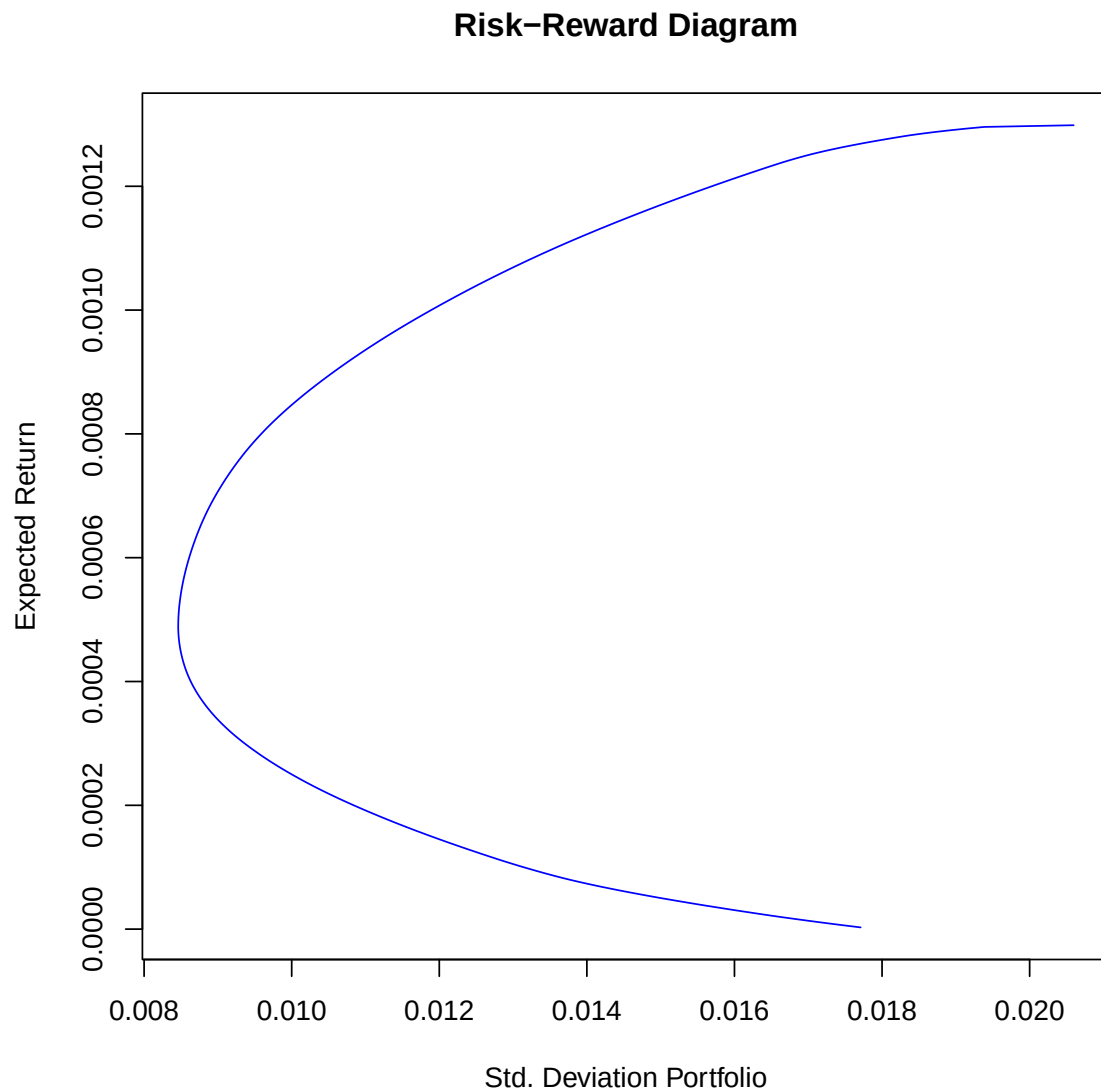
(c) Repeat part (b) with a suitably chosen range of target expected returns under the condition that no short-selling is allowed (i.e., the weights are all non-negative). Comment on the optimal portfolio weights in the two cases, i.e., with and without short-selling. Is the expected return in part (b) sustainable/reasonable?

```
v.range = seq(from=0.001,to=0.75,by=0.001)/365;
v.range = v.range[v.range<max(mu)]
sig.pf = matrix(0,nrow=length(v.range),ncol=1);
weights = matrix(0,nrow=length(v.range),ncol=30);
for (i in (1:length(v.range))) {
```

```

v = v.range[i];
res = run_solve_QF(Sigma, mu, v, short_sell=T)
sig.pf[i] = sqrt(res$variance);
weights[i,] = res$solution;
}
mu_f = log(1.03)/365; # Risk-free rate
plot(sig.pf, v.range, type='l', col='blue',
      xlab="Std. Deviation Portfolio", ylab='Expected Return',
      main = "Risk-Reward Diagram")

```




```
sharp.ratio = (v.range - mu_f)/sig.pf
opt.sr.idx = which.max(sharp.ratio)
optimal.pf.long = c(v.range[opt.sr.idx], sig.pf[opt.sr.idx])
library(knitr)
kable(weights[opt.sr.idx, ])
```

x
0.0322195
0.0000000
0.0000000
0.0000000
0.0000000
0.1114488
0.0437215
0.0000000
0.0000000
0.0685954
0.0803847
0.0000000
0.0000000
0.0809055
0.0374103
0.0901444
0.0000000
0.0000000
0.0323894
0.0820323
0.0023124
0.0691579
0.0915812
0.0000000
0.0035324
0.0000000
0.0566435
0.1175210
0.0000000
0.0000000

```
kable(rbind(optimal.pf.short, optimal.pf.long),
      col.names = c("Expected Return", "Std. Deviation"))
```

	Expected Return	Std. Deviation
optimal.pf.short	0.0017562	0.0180899
optimal.pf.long	0.0009397	0.0110521

In the case with short-selling, the expected return is 0.17% and the standard deviation is 1.8%. In the case without short-selling, the expected return is 0.09% and the standard deviation is 1.1%.

We can notice that the case that allows short-selling almost doubles the expected return of the other case. If we observe the weights of the optimal portfolio in (b), we note that the third stock has a strong short position of 33%. Thus, short-selling allows the portfolio to invest more resources on high-yield stocks, and at the same time, it can profit from stocks that have losses. In conclusion, the expected return of portfolio (b) is only sustainable while there are companies experiencing losses.

3. Problem 6 on page 513 of the textbook ([link to the textbook](#)).

(a) The value for the beta of stock A can be calculated from:

$$\begin{aligned}\beta_A &= \frac{\sigma_{AM}}{\sigma_M^2} \\ &= \frac{165\%^2}{11^2\%^2} \\ &= 1.364\end{aligned}$$

This value of beta implies an "agressive" stock.

(b) The expected return can be calculated from:

$$\begin{aligned}\mu_A &= \mu_f + \beta_A(\mu_M - \mu_f) \\ &= 4\% + 1.364 \cdot (12\% - 4\%) \\ &= 14.91\%\end{aligned}$$

(c) Finally, the percentage of the variance due to market risk is:

$$\begin{aligned}\frac{\beta_A^2 \sigma_M^2}{\sigma_A^2} &= \frac{1.364^2 11^2\%^2}{320\%^2} \\ &= 0.7031\end{aligned}$$

In conclusion, the percentage of variance due to market risk is 70.31%.

4. Problem 7 on page 513 of the textbook.

(a) The beta of the portfolio is the average of the given betas.

$$\begin{aligned}\beta_P &= \frac{\beta_1 + \beta_2 + \beta_3}{3} \\ &= 0.8667\end{aligned}$$

(b) The requested variance satisfies:

$$\begin{aligned}\text{Var}(R_P - \mu_{ft}) &= (\beta_P)^2 \text{Var}(R_{Mt} - \mu_{ft}) + \frac{1}{3} \sum_j \text{Var}(\epsilon_j) \\ &= 0.8667^2 \cdot 0.014 + \frac{0.010 + 0.015 + 0.011}{3} \\ &= 0.02252\end{aligned}$$

(c) We calculate the percentage as:

$$\frac{(\beta_P)^2 \text{Var}(R_{Mt} - \mu_{ft})}{0.2252} = 0.4670$$

Thus, the percentage of the risk due to market risk is 46.7%.

5. Problem 2 on page 356 in the textbook (read Problem 1 therein on how to get the data).

```
library(Ecdat)

## Warning: package 'Ecdat' was built under R version 4.4.3
## Loading required package: Ecfun
## Warning: package 'Ecfun' was built under R version 4.4.3
##
## Attaching package: 'Ecfun'
## The following object is masked from 'package:base':
##
##     sign
##
## Attaching package: 'Ecdat'
## The following object is masked from 'package:datasets':
##
##     Orange

data(CRSPday)
crsp = CRSPday[,7]

ar1 = arima(crsp, order=c(1,0,0))
ar2 = arima(crsp, order=c(2,0,0))
selection = data.frame(
  Model = c("AR(1)", "AR(2)"),
  AIC = c(ar1$aic, ar2$aic),
  BIC = c(BIC(ar1), BIC(ar2))
)
library(knitr)
kable(selection)
```

Model	AIC	BIC
AR(1)	-17406.37	-17388.86
AR(2)	-17404.87	-17381.53

We can observe that both AIC and BIC are lower for the model AR(1) compared to AR(2). Then, AR(1) is preferred.

(b) We calculate the 95%-confidence interval from the previous model.

```

phi.hat = as.numeric(ar1$coef[1])
se.phi = sqrt(ar1$var.coef[1,1])
ci = data.frame(
  Parameter = c("phi"),
  Estimate = c(phi.hat),
  LowerBound = c(phi.hat-1.96*se.phi),
  UpperBound = c(phi.hat+1.96*se.phi)
)
kable(ci)

```

Parameter	Estimate	LowerBound	UpperBound
phi	0.0853018	0.0464615	0.1241421

Then, the 95% CI for ϕ is $[0.0465, 0.1241]$.

6. Problem 4 on page 356 in the textbook.

(a) The variance of the Y_t process is independent of time. It can be calculated from:

$$\begin{aligned}
 \text{Var}(Y_1) &= \gamma(0) = \frac{\sigma_\epsilon^2}{1 - \phi^2} \\
 &= \frac{1.2}{1 - 0.4^2} = 1.4286
 \end{aligned}$$

(b) Now, we calculate the covariances:

$$\begin{aligned}
 \text{Cov}(Y_1, Y_2) &= \gamma(1) = \phi \cdot \gamma(0) \\
 &= 0.4 \cdot 1.4286 = 0.5714 \\
 \text{Cov}(Y_1, Y_3) &= \gamma(2) = \phi^2 \cdot \gamma(0) \\
 &= 0.4^2 \cdot 1.4286 = 0.2286
 \end{aligned}$$

(c) Finally, we calculate the requested variance:

$$\begin{aligned}
 \text{Var}(0.5(Y_1 + Y_2 + Y_3)) &= 0.5^2 \text{Cov}(Y_1 + Y_2 + Y_3, Y_1 + Y_2 + Y_3) \\
 &= 0.25 [\text{Var}(Y_1) + \text{Var}(Y_2) + \text{Var}(Y_3) \\
 &\quad + 2(\text{Cov}(Y_1, Y_3) + \text{Cov}(Y_1, Y_2) + \text{Cov}(Y_2, Y_3))] \\
 &= 0.25 [3\gamma(0) + 2(\gamma(2) + 2\gamma(1))] \\
 &= 0.25 [3\gamma(0) + 2\gamma(2) + 4\gamma(1)] \\
 &= 1.7571
 \end{aligned}$$